Reflective Diary - Entry 1

Luke Camilleri | ARI3217 | Task 19 | 8 November 2026
Draft for student review | 503 words (citations, tables, figures and references excluded)

I repackaged the supplied LightGBM baseline without changing its method, using configuration, shared preprocessing, tests and a thin notebook. On the January-August 2017 holdout, AUPRC was 0.786 for City and 0.628 for Resort. The booking pipeline took 1.80 seconds, including loading, feature construction, training and scoring; this is not an isolated fit-time measurement. A separate fresh clone reproduced the baseline with pinned dependencies. [E1-E2]

Data inspection showed why reproducing scores is insufficient. There were 31,994 extra exact duplicate rows, 1,959 zero-ADR records and a large nationality imbalance between cancelled and arrived offline bookings (Table 2). These findings challenge random splits, naive revenue calculations and nationality-based explanations. My additional check found 27 Check-Out records whose status date disagreed with arrival plus recorded nights. This does not prove corruption: the source permits nights stayed or booked-to-stay. I added an automated audit rather than silently relabelling or deleting them. [E3; R1]

The Docker test challenged my assumption that passing local tests meant the project was portable. The container initially lacked PyArrow, which the saved model needed when loading pandas metadata. Testing in Ubuntu exposed a dependency that the Mac environment had concealed. The corrected image matched host batch predictions within the recorded tolerance. [E2]

The most significant finding was that technical success did not establish predictive reliability. Both load phases returned all 600 measured requests successfully, with API p95 below 35 ms, yet the Resort model failed its calibration check: its expected/observed cancellation ratio was 0.68 against the 0.85-1.15 band. I learnt to distinguish service monitoring from model monitoring: a healthy dashboard cannot immediately establish prediction quality when outcomes arrive later. The quality evidence here is historical replay, not live hotel outcomes. [E2]

I was initially confused about manager approval: did it mean signing a document or approving recommendations inside the system? This helped me distinguish approval of the overall policy from approval of a particular night's quantity. We implemented a simulated nightly review that records the reviewer, quantity and reason, while recognising that it does not replace policy-wide sign-off or authorise real hotel operations. [E4]

Mapping business, data, application and technology clarified these responsibilities (Figure 1). I would retain human approval because capacity is unknown and calibration remains imperfect, although human review itself does not guarantee better decisions. Automatic acceptance could remove review delay, but its end-to-end latency and running cost are unmeasured. With illustrative review labour of EUR2.50/day, one extra EUR250 relocation every 100 days would erase that saving. I would reconsider automation after verifying capacity, independently evaluating policy outcomes and measuring review delays and costs. [E5-E6]

My working AI Act assessment for ordinary commercial overbooking advice is non-high-risk under Article 6 and Annex III. The provider would be the entity supplying it under its name; the hotel operator would be the deployer. The local coursework test requires a separate Article 2(8) scope assessment. A changed intended use would require reassessment. [E7; R2-R5]

I used Codex to assist with implementation, tests, data investigation, diagrams and drafting. This assistance makes it especially important that I can explain the results and justify the decisions myself.

Evidence: reproduction and operation

Supporting material - excluded from the reflective word count. Values describe the tested experiments, not production guarantees.

Table 1. Reproduction, quality and service measurements

Measure	Result and boundary	Evidence
Baseline holdout	40,687 bookings; Jan-Aug 2017. City AUPRC 0.785722; Resort 0.627908.	E1-E2
Calibration diagnostic	Expected/observed: City 0.911281 PASS; Resort 0.681723 FAIL. Reference band 0.85-1.15.	E2
Booking pipeline	1.798645 seconds, including loading, features, fit and scoring; 20/20 reference comparisons passed.	E2
Fresh clone	Separate pinned environment; 16 tests, 20 baseline comparisons and thin notebook passed. Code tested: 39fb510.	E1
Load: booking phase	300 measured requests; 8 workers; offered 5 requests/s for 60 s. p50 29.59 ms; p95 34.25 ms; p99 36.13 ms; 0 errors.	E2
Load: shifted phase	300 measured requests; same load; synthetic lead-time shift. p50 29.36 ms; p95 33.62 ms; p99 36.80 ms; 0 errors.	E2
Load interpretation	20 warmups excluded in total. Client timing includes scheduled-arrival queueing. All 600 measured prediction IDs persisted. Target p95 <250 ms; not a maximum-capacity test.	E2
Batch example	81 bookings, 44.72 ms including model loading/inference; excludes full database, PMS and human workflow.	E2
Container and parity	212,353,366 bytes (212.35 MB); stop 0.187 s, exit 0. Host/container batch match: absolute and relative tolerances 1e-12.	E2
Linux verification	118 tests, no failures/errors/skips; dashboard 8 panels; 6 healthy alert-rule evaluations. External alert delivery is not configured.	E2

Table 2. Listed quirks and an additional independent audit

Observation	Count / implication
Extra exact duplicate rows	31,994 / 119,390 rows; random-split overlap can inflate evaluation.
Zero ADR	1,959 rows; zero quoted price must not silently become ordinary room revenue.
Nationality/channel artefact	Within Offline TA/TO: PRT accounts for 8,121/8,311 cancellations (97.71%) and 3,864/15,908 arrivals (24.29%). These are nationality shares, not cancellation rates; causation is unproven.
Additional: Check-Out date mismatch	27 / 75,166 Check-Out rows (21 Resort, 6 City; 24 distinct rows). Status date differs from arrival plus recorded nights. Counted by a tested audit; interpretation remains uncertain.

Sources: E3 (committed audit and underlying quirk counts); dataset semantics R1. The unchanged baseline is retained separately from diagnostic audits.

Architecture and capability choice

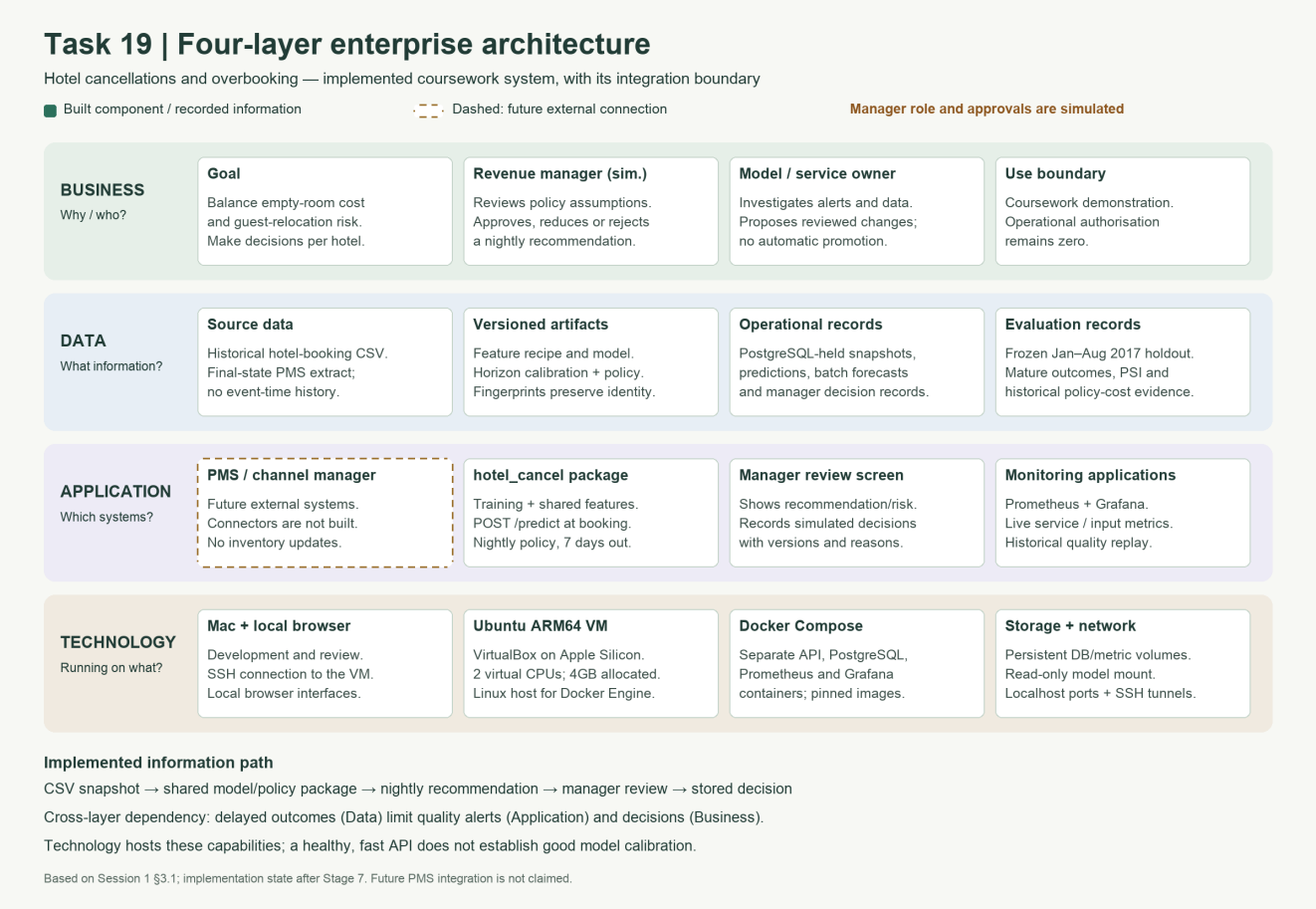


Figure 1. Four enterprise-architecture layers and the delayed-outcome feedback path. Solid components are implemented; future external integrations are marked in the figure. Source: E5.

Table 3. Human review versus hypothetical automatic acceptance

Criterion	Implemented: human review	Hypothetical: automatic acceptance
Latency budget	Measured API p95 34.25ms; example batch 44.72ms; human turnaround unmeasured	Common calculation reused in proposal; human wait removed; connector/end-to-end latency unmeasured
Running cost/day	Shared platform unmeasured + assumed EUR2.50/day review labour + oversight and policy losses	Shared platform unmeasured; potentially saves assumed EUR2.50/day review labour; added control/integration/incident costs unknown
Decision owner	Simulated manager records approve/reduce/reject; project owner maintains model/service	System would execute an authorised policy; business owner remains accountable; not implemented
Failure behaviour	No approval means no approved quantity; database/review failure blocks a new decision; operational fallback zero/manual	Would require fail-closed capacity/quality/version/connector checks; these automatic controls are not built

Cost assumption: two hotels x 5 minutes/day x EUR15/hour = EUR2.50/day. This is illustrative labour, not measured total operating cost. Automatic acceptance and its safeguards are not implemented. Source: E6.

Monitoring and source register

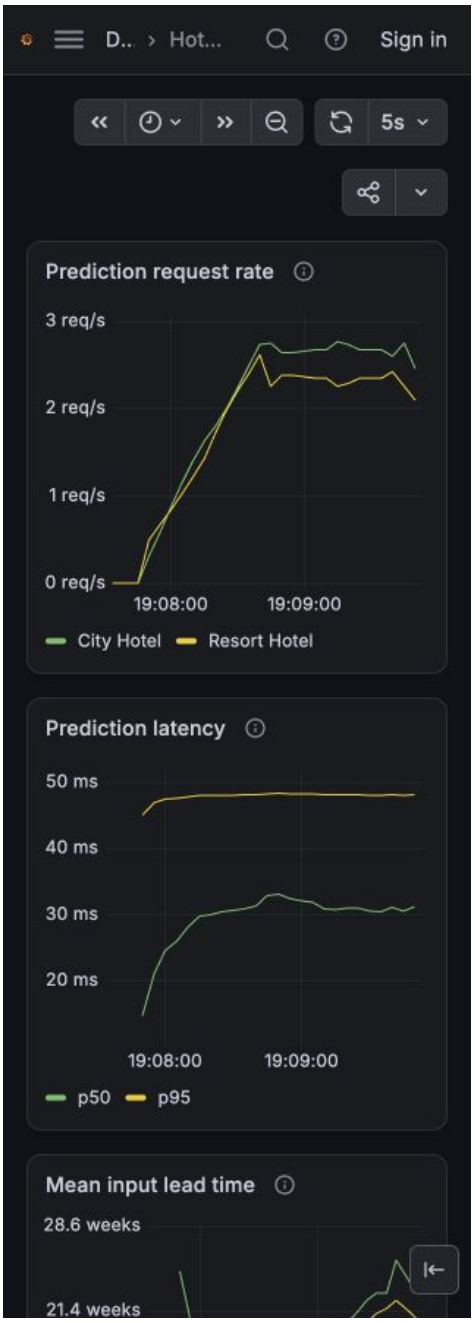


Figure 2. Saved Grafana preview from the Stage 7 experiment. Request rate, latency and input panels support service observation; they do not establish immediate cancellation quality. Historical replay supplies quality evidence separately. Dashboard latency is estimated from server histogram buckets; Table 1 uses client timings, so their percentiles differ. Source: E2.

Committed project evidence

Paths below are relative to the hotel-cancellations repository. Commit IDs identify the committed evidence; generation metadata may name an earlier base commit.

E1 e3605b4 - docs/fresh-clone-verification.md and docs/evidence/fresh-clone/. Fresh-clone code: 39fb510.

E2 6ccc11e - docs/stage7-monitoring.md; docs/evidence/stage7/{baseline-regression,performance,batch,container,tests-linux,verification}.json. Screenshot: artifacts/stage7/dashboard-preview.jpg (generated, not version-controlled).

E3 e57de9a - docs/diary1-additional-quirk.md; docs/evidence/diary1/additional-quirk.json; docs/evidence/data-quirks.json; tests/test_stay_audit.py.

E4 bb8d11d - docs/manager-approval.md and simulated approval implementation. Policy-wide sign-off remains separate.

E5 1b5cbc9 - docs/diary1-architecture.md; docs/figures/diary1-four-layer-architecture.png.

E6 dec3510 - docs/diary1-pattern-comparison.md; docs/evidence/diary1/pattern-comparison*.csv and pattern-cost-sensitivity.csv.

E7 e705484 - docs/diary1-ai-act-classification.md; docs/evidence/diary1/ai-act-classification.json. Scoped working assessment, reviewed 8 November 2026.

Primary references

R1 Antonio, de Almeida & Nunes (2019), Hotel booking demand datasets. Data in Brief 22, 41-49.

R2 European Commission AI Act Explorer: Article 6 - high-risk classification.

R3 European Commission AI Act Explorer: Annex III - listed uses.

R4 European Commission AI Act Explorer: Article 3 - provider/deployer definitions.

R5 European Commission AI Act Explorer: Article 2(8) - pre-deployment testing scope.

Course framing: ARI3217 Reflective diary brief; Session 1, sections 3.1 and 3.3. Review the draft in your own words before submission. Slides and the own-voice video are separate deliverables.